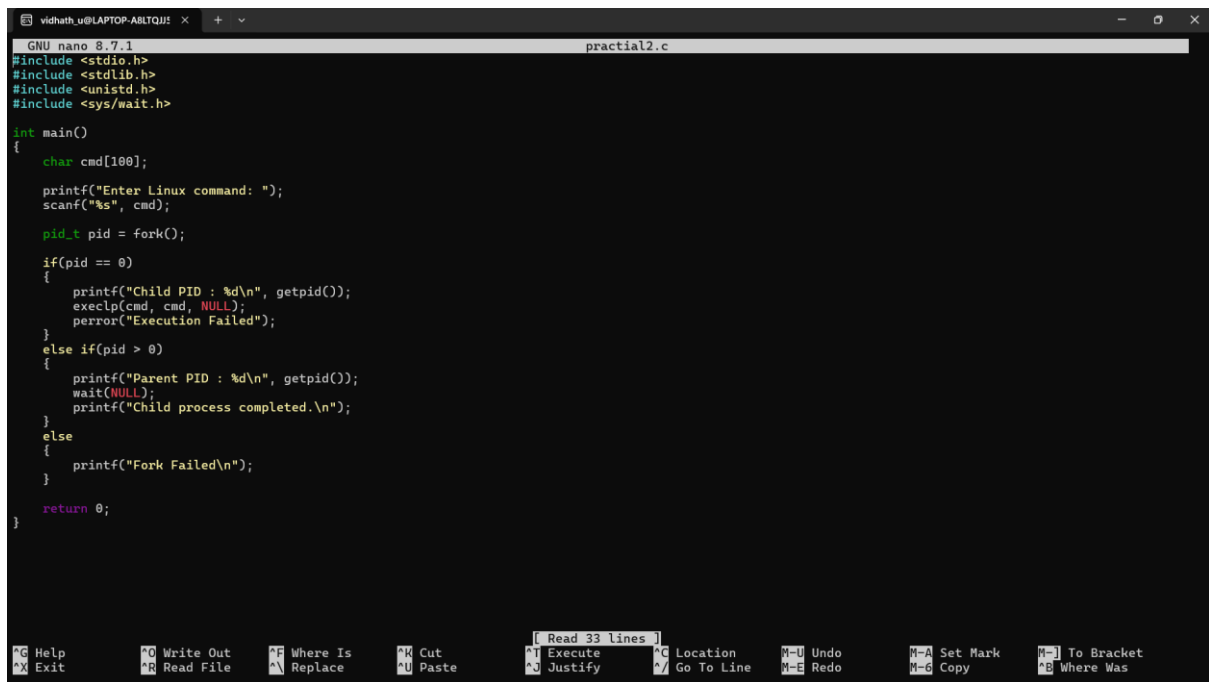


1.



The screenshot shows a terminal window with the GNU nano 8.7.1 editor open. The file being edited is named 'practial2.c'. The code is a C program that demonstrates the use of the fork() system call to create a child process. The program includes headers for stdio, stdlib,unistd, and sys/wait. The main function prompts the user to enter a Linux command, which is then executed in a child process using execlp(). The parent process waits for the child to complete and prints the child's PID and the parent's PID. The program also includes error handling for fork and execution failures.

```
GNU nano 8.7.1 practial2.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];

    printf("Enter Linux command: ");
    scanf("%s", cmd);

    pid_t pid = fork();

    if(pid == 0)
    {
        printf("Child PID : %d\n", getpid());
        execlp(cmd, cmd, NULL);
        perror("Execution Failed");
    }
    else if(pid > 0)
    {
        printf("Parent PID : %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }
    else
    {
        printf("Fork Failed\n");
    }

    return 0;
}
```

The bottom of the window shows the nano editor's status bar with various keyboard shortcuts for editing and navigation.

```
vidhath_u@LAPTOP-A8LTQJJ5:~$ nano practial2.c
vidhath_u@LAPTOP-A8LTQJJ5:~$ gcc practical2.c -o practical2
cc1: fatal error: practical2.c: No such file or directory
compilation terminated.
vidhath_u@LAPTOP-A8LTQJJ5:~$ gcc practical2.c -o practical2
cc1: fatal error: practical2.c: No such file or directory
compilation terminated.
vidhath_u@LAPTOP-A8LTQJJ5:~$ gcc practial2.c -o practical2
vidhath_u@LAPTOP-A8LTQJJ5:~$ ./practical2
Enter Linux command: uname
Child PID : 567
Parent PID : 521
Linux
Child process completed.
```

```
vidhath_u@LAPTOP-A8LTQJJ5:~$ ./practical2
Enter Linux command: lscpu
Parent PID : 577
Child PID : 592
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      16
On-line CPU(s) list:         0-15
Vendor ID:                   GenuineIntel
Model name:                  13th Gen Intel(R) Core(TM) i7-13620H
CPU family:                  6
Model:                       186
Thread(s) per core:          2
Core(s) per socket:          8
Socket(s):                   1
Stepping:                    2
BogoMIPS:                    5836.80
Flags:                        fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                             r pge mca cmov pat pse36 clflush mmx fxsr sse s
                             se2 ss ht syscall nx pdpe1gb rdtscp lm constant
                             _tsc rep_good nopl xtopology tsc_reliable nonst
                             op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
                             sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                             opcnt tsc_deadline_timer aes xsave avx f16c rdr
                             and hypervisor lahf_lm abm 3dnowprefetch ssbd i
                             brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
                             d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi
                             2 erms invpcid rdseed adx smap clflushopt clwb
                             sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
```

```

Virtualization features:
  Virtualization:      VT-x
  Hypervisor vendor:   Microsoft
  Virtualization type: full
Caches (sum of all):
  L1d:                 384 KiB (8 instances)
  L1i:                 256 KiB (8 instances)
  L2:                  10 MiB (8 instances)
  L3:                  24 MiB (1 instance)
NUMA:
  NUMA node(s):        1
  NUMA node0 CPU(s):   0-15
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite:           Not affected
  Indirect target selection: Not affected
  Itlb multihit:        Not affected
  L1tf:                 Not affected
  Mds:                  Not affected
  Meltdown:             Not affected
  Mmio stale data:      Not affected
  Old microcode:        Not affected
  Reg file data sampling: Mitigation; Clear Register File
  Retbleed:             Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass:    Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:           Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:           Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
  Srbds:                Not affected
  Tsa:                  Not affected
  Tsx async abort:      Not affected
  Vmscape:              Not affected
Child process completed.

```

```
vidhath_u@LAPTOP-A8LTQJJ5:~$ ./practical2
Enter Linux command: ps
Parent PID : 602
Child PID : 603
```

```
  PID TTY          TIME CMD
  314 pts/0        00:00:00 bash
  602 pts/0        00:00:00 practical2
  603 pts/0        00:00:00 ps
```

Child process completed.

```
vidhath_u@LAPTOP-A8LTQJJ5:~$ ./practical2
Enter Linux command: top
Parent PID : 613
Child PID : 614
```

```
top - 09:42:28 up 11 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  27 total,   1 running, 26 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.3 us,   0.0 sy,   0.0 ni, 99.7 id,   0.0 wa,   0.0 hi,   0.0 si,   0.0 st
MiB Mem : 11775.6 total, 11182.8 free,   595.7 used,   167.0 buff/cache
MiB Swap: 3072.0 total, 3072.0 free,    0.0 used, 11179.9 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24180	15432	11612	S	0.0	0.1	0:00.46	systemd
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.00	init-systemd(Ub
6	root	20	0	3180	2040	1940	S	0.0	0.0	0:00.00	init
46	root	19	-1	50404	16760	15488	S	0.0	0.1	0:00.13	systemd-journal
79	systemd+	20	0	22416	14444	11952	S	0.0	0.1	0:00.05	systemd-resolve
84	root	20	0	35468	12384	9188	S	0.0	0.1	0:01.55	systemd-udev
104	root	20	0	2888	1948	1820	S	0.0	0.0	0:00.02	chronyd-starter
105	root	20	0	4472	3108	2860	S	0.0	0.0	0:00.00	cron
106	message+	20	0	8796	5368	4680	S	0.0	0.0	0:00.03	dbus-daemon
110	root	20	0	44096	29804	14668	S	0.0	0.2	0:00.15	networkd-dispat
115	root	20	0	18436	9392	8148	S	0.0	0.1	0:00.07	systemd-logind
186	syslog	20	0	220548	5896	4604	S	0.0	0.0	0:00.04	rsyslogd
198	root	20	0	5492	2732	2548	S	0.0	0.0	0:00.00	agetty
211	_chrony	20	0	23592	11188	7204	S	0.0	0.1	0:00.06	chronyd
229	_chrony	20	0	12096	2404	1760	S	0.0	0.0	0:00.00	chronyd
245	root	20	0	123460	32728	18120	S	0.0	0.3	0:00.11	unattended-upgr
306	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00	SessionLeader
308	root	20	0	3200	1252	1108	S	0.0	0.0	0:00.05	Relay(314)
314	vidhath+	20	0	6268	5600	3928	S	0.0	0.0	0:00.10	bash
316	root	20	0	8644	5600	4760	S	0.0	0.0	0:00.00	login